

9(a)

any **seven** from:

- 1 calcium ions released from sarcoplasmic reticulum ;
- 2 calcium ions bind to troponin ;
- 3 troponin changes shape and moves tropomyosin ;
- 4 exposes binding site on actin ;
- 5 myosin head, binds to site / forms cross bridge ;
- 6 myosin head tilts ;
- 7 pulls actin / power stroke ;
- 8 myosin head, has ATPase / hydrolyses ATP ;
- 9 myosin head lets go of actin ;
- 10 myosin head goes back to previous orientation / myosin head re-cocks ;
- 11 process repeated ;
- 12 sarcomere shortens ;

7

Question	Answer	Marks
9(b)	<i>any eight from:</i> 1 FSH secreted by anterior pituitary ; 2 stimulates, development / growth, of follicle (cells in ovary) ; 3 dominant / Graafian, follicle, secretes oestrogen ; 4 oestrogen stimulates repair of endometrium ; 5 oestrogen inhibits further release of FSH ; 6 (large) increase oestrogen, day 14 / midpoint ; 7 stimulates secretion of LH from anterior pituitary ; 8 LH stimulates, ovulation / release of oocyte ; 9 LH stimulates development of corpus luteum ; 10 corpus luteum secretes progesterone ; 11 progesterone continues build-up of endometrium or maintains endometrium ; 12 progesterone, inhibits secretion of, LH / FSH ; 13 corpus luteum degenerates so concentration of progesterone falls ; 14 endometrium breaks down ;	8